

RATNAPURA, Sri Lanka

WMO: 434860

Lat: 6.7129N

Lon: 80.3808E

Elev: 289

StdP: 14.54

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	69.8	71.4	64.3	91.3	81.1	66.7	99.3	80.8	11.7	80.4	9.7	81.6	0.4	N/A	0.265

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.8	93.3	78.1	92.0	78.2	90.9	78.1	81.3	88.8	80.7	88.0	80.1	87.0	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	153.8	84.5	78.9	151.5	84.1	78.3	148.6	83.4	45.3	89.2	44.6	88.2	44.1	87.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.0	8.1	7.0	DB	62.6	97.3	4.7	3.0	59.1	99.5	56.4	101.3	53.7	102.9	50.3	105.1
			WB	62.4	82.8	4.4	0.7	59.3	83.3	56.8	83.7	54.3	84.1	51.1	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	80.6	81.7	82.8	82.7	82.1	80.9	81.1	80.5	80.4	80.5	80.3	79.8
	DBStd	2.13	2.26	1.82	1.76	1.88	2.03	1.86	1.88	1.73	1.91	1.78	1.65	2.20
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11355	947	888	1017	982	995	928	963	944	911	946	910	924
	CDD65	5880	482	468	552	532	530	478	498	479	461	481	460	459
	CDH74	50384	3944	4058	5025	4729	5174	4488	4740	4188	3794	3630	3247	3367
(13)	CDH80	16591	1522	1668	2134	1851	1712	1177	1292	1061	1026	1073	1002	1073
(14)	Wind	WSAvg	1.8	1.3	1.4	1.6	1.5	1.8	2.8	3.1	2.7	2.0	1.7	1.1
(15) Precipitation	PrecAvg	146.3	4.1	5.0	8.1	13.7	19.5	15.4	12.1	11.7	16.6	17.3	14.0	8.9
	PrecMax	184.1	8.3	12.7	12.2	22.5	38.8	33.2	25.3	21.6	30.0	27.0	26.4	14.5
	PrecMin	118.3	0.0	1.1	4.8	6.2	6.3	6.5	3.7	4.2	6.1	8.5	2.6	2.9
	PrecStd	16.5	2.2	2.8	2.2	3.8	8.5	5.7	6.0	4.8	7.3	4.5	5.5	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	94.4	95.3	94.5	92.7	90.0	90.6	89.8	89.9	91.1	90.9	90.9
		MCWB	75.8	76.5	77.3	79.8	80.3	79.4	78.9	78.2	78.4	78.7	78.3	77.2
	2%	DB	90.9	92.3	93.3	92.6	91.0	88.2	88.7	88.3	88.6	89.4	89.1	89.1
		MCWB	75.1	76.5	77.7	80.0	80.2	79.1	78.2	78.1	77.9	78.4	78.2	76.7
	5%	DB	89.3	90.8	91.8	91.1	89.5	87.0	87.3	87.0	87.4	87.9	87.6	87.7
		MCWB	75.5	76.3	77.9	80.1	80.0	78.9	77.9	77.9	77.8	78.2	78.2	76.8
	10%	DB	87.8	89.1	90.2	89.4	87.8	85.7	85.9	85.4	85.8	86.0	86.0	86.1
		MCWB	75.8	76.5	77.9	79.7	79.8	78.6	77.6	77.7	77.7	77.8	78.0	76.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.5	80.1	80.8	82.3	82.1	81.1	80.3	80.1	80.1	80.5	80.1	80.0
		MCDB	87.7	88.8	90.0	91.2	89.3	86.7	87.3	85.7	86.2	88.3	87.5	87.2
	2%	WB	78.5	79.1	79.9	81.4	81.3	80.2	79.3	79.2	79.2	79.6	79.3	79.1
		MCDB	86.3	87.8	89.0	89.8	88.1	85.5	85.9	85.0	85.2	86.8	86.4	86.0
	5%	WB	77.7	78.3	79.2	80.7	80.7	79.7	78.7	78.7	78.7	78.9	78.7	78.2
		MCDB	85.0	86.5	87.9	88.8	87.2	84.9	85.1	84.5	84.5	85.6	85.6	84.6
	10%	WB	77.0	77.5	78.6	79.9	80.1	79.1	78.1	78.1	78.1	78.2	78.1	77.5
		MCDB	83.8	85.2	86.9	87.2	85.9	84.1	83.9	83.5	83.5	84.0	84.1	83.4
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.3	17.4	15.5	11.8	9.5	10.1	10.2	11.2	12.6	13.2	13.8
		MCDBR	19.0	19.6	19.1	17.5	14.7	11.5	12.2	12.9	13.7	15.1	15.3	16.5
		MCWBR	5.6	5.7	5.6	6.0	5.3	4.4	4.3	4.6	5.0	5.5	5.6	5.8
	5% WB	MCDBR	15.6	17.0	17.4	16.0	12.6	10.5	10.7	10.8	11.9	13.4	13.8	14.4
		MCWBR	5.6	5.8	6.0	6.3	5.2	4.8	4.7	4.7	5.1	5.6	5.6	5.9
(40) Clear-Sky Solar Irradiance	taub		0.432	0.452	0.492	0.498	0.512	0.493	0.510	0.491	0.462	0.470	0.465	0.446
	taud		2.346	2.276	2.179	2.190	2.143	2.212	2.137	2.208	2.308	2.277	2.289	2.327
	Ebn at Noon		280	278	267	261	251	253	249	259	271	269	269	274
	Edh at Noon		40	44	49	48	49	45	49	47	43	44	43	41
(44) All-Sky Solar Radiation	RadAvg		1580	1786	1879	1778	1610	1577	1591	1614	1642	1573	1414	1402
	RadStd		120	105	108	95	122	96	71	80	114	94	95	125

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page